

Phan Anh VU (♂)

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AI and chip design research

Experience

PhD Researcher, CEA (Commissariat à l'Énergie Atomique et alternative)

 2023/03 - 2026/07  Grenoble, France

- Develop novel methods for efficient uncertainty quantification in deep learning, with a focus on hardware-software co-design. Implement the proposed algorithm in Jax, using Vision Transformer and MLP Mixer architecture, and benchmark against existing methods.
- Implement the proposed algorithm method in hardware. Design a novel analog in-memory computing circuit using memristor with Cadence Virtuoso. Simulate the circuit with a CMOS technology process using Eldo.

Intern Researcher, ENS Paris Saclay SATIE Lab

 2022/02 - 2022/08  Paris, France

Study the prospects of using probabilistic and physics-informed machine learning for predictive maintenance of power electronics with time series data. Benchmark PyTorch implementations of various methods for uncertainty quantification in deep learning.

Intern Researcher, Université Paris Saclay LISN Lab

 2021/04 - 2021/09  Paris, France

Benchmark PyTorch implementations of various methods for uncertainty quantification in deep learning.

Software Engineer, Cegid Talentsoft

 2017/11 - 2019/12  Paris, France

- For a talent management web application with estimated 9 million users and 2000 institutional clients, diagnose and resolve complex critical bugs, of various aspects such as front-end (JavaScript and TypeScript), back-end (C#, .NET), database (SQL Server), infrastructure.
- Implement new features, developer tools.

Skills

Machine Learning & Data Science: Python, Jax, PyTorch, Keras, TensorFlow, numpy, pandas, scikit-learn, matplotlib, seaborn, Julia

Electronic Design Automation (EDA): Cadence Virtuoso, Eldo, VHDL

Software Development: C#, .NET ASP MVC, JavaScript, TypeScript, Git, SQL, Redis

Publications

Adaptive Low-Rank Ensemble: estimation of ensemble performance as a function of the size, and reduce memory size by 3 up to 6 with equal performance as traditional ensemble (patent pending, preprint available on request).

Compressed vector-matrix multiplication for Memristor-based ensemble neural networks (patented), *IEEE International Conference on Rebooting Computing 2024*: an analog in-memory computing circuit with memristor to implement vector element-wise multiplication and low-rank approximation of ensemble of neural networks ([Blog page](#)).

Probabilistic and Physics-Informed Machine Learning for Predictive Maintenance with Time Series Data, *EuroSimE 2023*: ([Code repository](#))

Meta-Album: Multi-domain Meta-Dataset for Few-Shot Image Classification, *NeurIPS 2022 Datasets and Benchmarks*: ([Website](#))

Education

PhD, Machine Learning & Electronics, Université Grenoble Alpes

 2023 - 2026  Grenoble, France

Master, Computer Science & Artificial Intelligence, Université Paris Saclay

 2020 - 2022  Paris, France

Languages

English, Français, Tiếng Việt
中文, Español, Deutsch

